LAPORAN PRAKTIKUM PERANCANGAN JARINGAN

Pemilihan dan Penerapan Protokol Switching dan Routing #3
- Membuat Topologi Pada Cisco Packet Tracer -

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1. Tujuan

a. Memahami perancangan jaringan secara *top-down*

b. Menerapkan rancangan jaringan berhirarki

c. Melakukan konfigurasi protokol-protokol

- VLAN
- VLAN dan penomoran ports VLAN
- Subnetting dan Pengalamatan IP
- Inter-VLAN Router pada Multilayer switches (Switch Virtual Interface)
- DHCP
- SSH
- OSPF
- Switching Security
- WLAN atau Wireless Network (Cisco Access Point)

d. Pengujian dan Verifikasi komunikasi jaringan

2. Dasar Teori

Faktor-faktor yang dilibatkan dalam sebuah pengambilan keputusan :

- Tujuan harus jelas
- Pilihan-pilihan atau ide yang ada harus dipelajari lebih jauh terlebih dahulu.
- Konsekuensi dari setiap keputusan harus dipelajari lebih lanjut
- Rencana cadangan harus segera dibuat.

Selain faktor-faktor diatas juga harus melibatkan beberapa pertanyaan berikut ini :

- Jika hasil keputusan yang diambil ternyata salah, apa yang akan dikerjakan ?
- Jika hasil keputusan sudah pernah dilakukan sebelumnya, apakah ada permasalahan dengan keputusan tersebut ?
- Bagaimana reaksi pelanggan dengan keputusan anda ?
- Apa rencana cadangan anda jika keputusan anda ditolak oleh pelanggan ?

Berikut ini adalah salah satu contoh dokumen dalam pembuatan keputusan

	Critical Goals			Other Goals		
	Adaptability—must adapt to changes in a large internetwork within seconds	Must scale to a large size (hundreds of routers)	Must be an industry standard and compatible with existing equipment	Should not create a lot of traffic	Should run on inexpensive routers	Should be easy to configure and manage
BGP	X*	X	X	8	7	7
OSPF	X	X	X	8	8	8
IS-IS	X	X	X	8	6	6
IGRP	X	X				
EIGRP	X	X				
RIP			X			

*X = Meets critical criteria. 1 = Lowest. 10 = Highest.

Pemilihan Switching Protocols

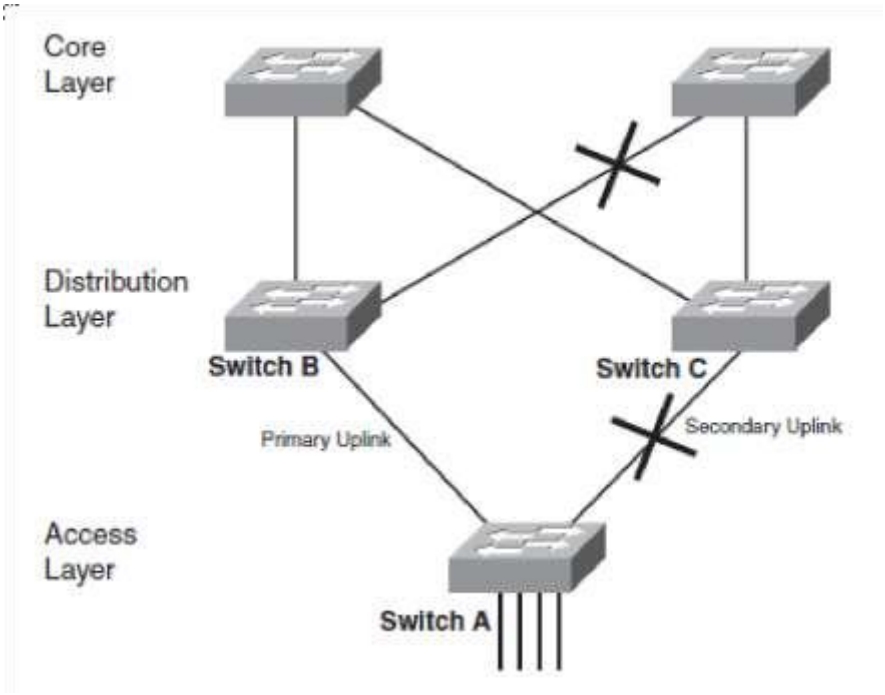
Switch memanfaatkan sirkuit terintegrasi cepat untuk menawarkan latensi rendah. Bridges jauh lebih lambat daripada switch dan memiliki lebih sedikit port dan biaya per port yang lebih tinggi. Untuk alasan ini, switch telah menggantikan bridges. Switch memiliki kemampuan untuk melakukan pemrosesan *store-and-forward* atau pemrosesan *cut-through*. Dengan pemrosesan *cut-through*, sebuah switch dengan cepat melihat alamat tujuan (bidang pertama dalam header frame Ethernet), menentukan port keluar, dan segera mulai mengirim bit ke port keluar. Kerugian dengan pemrosesan *cut-through* adalah meneruskan frame ilegal (misalnya, kerdil Ethernet) dan frame dengan kesalahan CRC.

Pada jaringan yang rawan kegagalan dan kesalahan, pemrosesan *cut-through* tidak boleh digunakan. Beberapa switch memiliki kemampuan untuk secara otomatis berpindah dari mode *cut-through* ke mode *store-and forward* ketika ambang kesalahan tercapai. Fitur ini disebut pengalihan *cut-through adaptif* oleh beberapa vendor. Switch memungkinkan beberapa jalur penerusan paralel, yang berarti sebuah switch dapat menangani volume lalu lintas yang tinggi lebih cepat daripada sebuah bridge. Switch high-end dapat mendukung banyak jalur penerusan simultan, tergantung pada struktur switching.

Berikut ini beberapa protokol switching yang dapat digunakan :

- STP
- RSTP
- PVST
- PVST+
- Etherchannel
- VTP
- DTP

Berikut ini contoh penggunaan STP



Gambar 1 Switch lapisan akses dengan dua uplinks ke switches layer distribusi

Pemilihan Routing Protocols

Protokol routing memungkinkan router secara dinamis mempelajari cara menjangkau jaringan lain dan bertukar informasi ini dengan router atau host lain. Memilih protokol routing untuk pelanggan desain jaringan Anda lebih sulit daripada memilih protokol switching, karena ada begitu banyak pilihan. Keputusan akan lebih mudah jika Anda dapat menggunakan tabel keputusan, seperti yang ditunjukkan pada Tabel 7-1 pada buku *Top-Down Network Design 3rd Edition*. Berbekal pemahaman yang kuat tentang tujuan dan informasi pelanggan Anda tentang karakteristik protokol routing yang berbeda, Anda dapat membuat keputusan yang tepat tentang

protokol routing mana yang direkomendasikan. Berikut ini tabel ringkasan panduan dalam pemilihan protocol routing

Tabel 1 Perbandingan protokol-protokol routing

	Distance Vector or Link State	Interior or Exterior	Classful or Classless	Metrics Supported	Scalability	Convergence Time	Resource Consumption	Supports Security? Authenticates Routes?	Ease of Design, Configuration, and Troubleshooting
RIPv1	Distance vector	Interior	Classful	Hop count	15 hops	Can be long (if no load balancing)	Memory: low CPU: low Bandwidth: high	No	Easy
RIPv2	Distance vector	Interior	Classless	Hop count	15 hops	Can be long (if no load balancing)	Memory: low CPU: low Bandwidth: high	Yes	Easy
IGRP	Distance vector	Interior	Classful	Bandwidth, delay, reliability, load	255 hops (default is 100)	Quick (uses triggered updates and poison reverse)	Memory: low CPU: low Bandwidth: high	No	Easy
EIGRP	Advanced distance vector	Interior	Classless	Bandwidth, delay, reliability, load	1000s of routers	Very quick (uses DUAL algorithm)	Memory: moderate CPU: low Bandwidth: low	Yes	Easy
OSPF	Link state	Interior	Classless	Cost (100 million divided by bandwidth on Cisco routers)	A few hundred routers per area, a few hundred areas	Quick (uses LSAs and Hello packets)	Memory: high CPU: high Bandwidth: low	Yes	Moderate
BGP	Path vector	Exterior	Classless	Value of path attributes and other configurable factors	1000s of routers	Quick (uses update and keepalive packets, and withdraws routes)	Memory: high CPU: high Bandwidth: low	Yes	Moderate
IS-IS	Link state	Interior	Classless	Configured path value, plus delay, expense, and errors	Hundreds of routers per area, a few hundred areas	Quick (uses LSAs)	Memory: high CPU: high Bandwidth: low	Yes	Moderate

Berikut juga adalah tabel administrative distance berdasarkan tipe route untuk preferensi penentuan rute oleh router.

Tabel 2 Administrative Distance berdasarkan tipe route

Route Source	Default Distance Value
Connected interface	0
Static route	1
Enhanced Interior Gateway Routing Protocol (EIGRP) summary route	5
External Border Gateway Protocol (BGP)	20
Internal EIGRP	90
IGRP	100
OSPF	110
Intermediate System-to-Intermediate System (IS-IS)	115
Routing Information Protocol (RIP)	120
Exterior Gateway Protocol (EGP)	140
On-Demand Routing (ODR)	160
External EIGRP	170
Internal BGP	200
Unknown	255

Simpulan Pemilihan Protokol

Pemilihan protokol switching dan routing yang tepat untuk pelanggan desain jaringan Anda sangat diperlukan. Skalabilitas dan karakteristik kinerja protokol dan seberapa cepat protokol dapat beradaptasi dengan perubahan menjadi perhatian penting dalam pemilihan protocol witching maupun routing.

Memutuskan protokol switching dan routing yang tepat untuk pelanggan Anda akan membantu Anda memilih produk switch dan router terbaik untuk pelanggan. Misalnya, jika Anda telah memutuskan bahwa desain harus mendukung protokol routing yang dapat menyatu dalam hitungan detik di internetwork besar, Anda mungkin tidak akan merekomendasikan router yang hanya menjalankan RIP.

Detail pembahasan bagaimana memilih protocol switching dan routing sangaat direkomendasikan untuk membaca buku Top-down Network Design 3rd Edition dari CISCO yang juga sebagai buku utama matakuliah teori Perancangan Jaringan. Buku tersebut terdapat diskusi umum tentang pengambilan keputusan untuk membantu

Anda mengembangkan proses sistematis untuk memilih solusi desain jaringan. Sebuah diskusi tentang menjembatani dan beralih protokol diikuti, yang meliputi menjembatani transparan, beralih multilayer, perangkat tambahan untuk protokol STP, dan VLAN. Bagian pada protokol routing mengikuti bagian switching.

3. Skenario Kasus pada pemanfaatan protokol-protokol jaringan sistem U Bank

Ikuti perintah-perintah dalam perancangan jaringan U Bank dan pengujian sesuai skenario berikut

Skenario Umum

Pada kasus sekenario Jaringan Sistem Bank ini terbagi menjadi dua bagian. Bagian pertama adalah perancangan topologi mengikuti *user requirement* dari klien menggunakan drawing tool. Bagian kedua mengimplementasikan desain topologi dan konfigurasi ke dalam simulator jaringan Packet Tracer. Ikuti Langkah-langkah pada modul.

U Bank adalah perusahaan milik Indonesia yang bergerak di bidang Perbankan dan Asuransi. Perusahaan tersebut bermaksud untuk memperluas layanannya di Asia Tenggara dengan memiliki cabang pertama yang berlokasi di Bangkok, Thailand. Perusahaan tersebut memiliki gedung empat lantai untuk beroperasi di ibu kota Thailand. Oleh karena itu, perusahaan ingin mengizinkan sumber pengetahuan dari sekelompok mahasiswa tahun terakhir dari Prodi Teknologi Rekayasa Internet SV UGM untuk merancang dan mengimplementasikan jaringan perusahaan mereka.

Asumsikan Anda adalah salah satu siswa yang mengambil alih peran ini, baca persyaratannya dengan cermat, lalu buat model desain dan implementasi jaringan berdasarkan kebutuhan perusahaan. Setiap lantai memiliki departemen seperti yang tertera pada Tabel 1-4 di bawah ini :

Tabel 1 Departemen di Lantai 1

Lantai Satu	
Departments	Devices
Management	PCs & Printers
Research	PCs & Printers
Human Resource	PCs & Printers

Tabel 2 Departemen di Lantai 2

Lantai Dua	
Departments	Devices
Marketing	PCs & Printers
Accounting	PCs & Printers

Finance	PCs & Printers

Table 3 Departemen di Lantai 3

Lantai Tiga	
Departments	Devices
Logistic	PCs & Printers
Customer Care	PCs & Printers
Guest Area	PCs & Printers

Table 4 Departemen di Lantai 4

Lantai Empat	
Departments	Devices
Administration	PCs & Printers
ICT	PCs & Printers
Server	Server

Untuk merancangnya, maka perlu dibuat diagram topologi jaringan menggunakan aplikasi untuk mempermudah visualisasi tim perancangan. Beberapa contoh aplikasi yang dapat digunakan untuk memvisualisasinya seperti Microsoft Visio, Visual Paradigm, atau Draw.io.

Adapun ketentuan dan kebutuhan dari perancangannya adalah sebagai berikut:

a. Menggunakan OSPF sebagai protokol routing
b. Pada Guest Area di lantai 3 terdapat jaringan wireless untuk users
c. Setiap departemen kecuali ruang server diantisipasi memiliki sekitar 60 pengguna jaringan
d. Perangkat host di jaringan memiliki alamat IPv4 secara otomatis
e. Perangkat di seluruh departemen diharapkan dapat saling berkomunikasi
f. Terdapat web server

g. Semua perangkat di dalam jaringan diharapkan mendapat IP address secara otomatis yang dialokasikan oleh DHCP server yang berlokasi di ruangan server
h. Mengkonfigurasi SSH di setiap router untuk remote login
i. Mengkonfigurasi dasar perangkat : hostname, line console, enable password, banner message, disable domain IP lookup, encrypt all configured password
j. Setiap departemen berada di VLAN dan subnetwork yang berbeda
k. Diberikan jaringan 192.168.10.0 sebagai alamat basis untuk jaringan. Lakukan subnetting berdasarkan jumlah host di setiap departemen sesuai yang diberikan. Identidikasi subnet mask, range IP address yang dapat digunakan, dan broadcast address untuk tiap subnet
l. Konfigurasi end-device: konfigurasi semua perangkat end-device dengan IP address yang sesuai dengan perhitungan di atas
m. Konfigutasi port-security: menggunakan sticky command untuk mendapatkan MAC Address dan Violation mode of the shutdown
n. Test dan verifikasi komunikasi jaringan
0. Nama hostname setiap device harus dengan format device_NIU ▪ misal Admin_SW_192151 / Lt2_SW_192151 dst. ▪ cek plagiasi, jika terbukti, maka nilai laporan 0.

4. Tugas Perancangan dan Desain Jaringan Sistem U Bank

Perancangan Tahap Kedua : Membuat Logical Topology di Cisco Packet Tracer untuk Simulasi

Langkah-langkah:

1. Buat topologi jaringan sesuai dengan bentuk diagram yang sudah dibuat pada draw.io. Sesuaikan perangkat dan cablingnya
2. Untuk menghubungkan cabling pada router, gunakan kabel serial, atur clock ratenya.
3. Untuk menghubungkan cabling pada router untuk backup linknya, gunakan kabel gigabit ethernet.
4. Nyalakan perangkat dan interface pada setiap perangkat. Untuk multi-layer switch, masukkan modul power supply pada perangkatnya.
5. Tambahkan access point pada departemen guest dan laptop sebagai end device uji cobanya.
6. Konfigurasi dasar perangkat. Sesuaikan penamaan perangkat mengikuti dengan penamaan yang sudah dibuat pada draw.io dengan format LANTAI-NAMA PERANGKAT-NIU MAHASISWA → misal Lt2_SW_192151 dst. → cek plagiasi, jika terbukti, maka nilai laporan 0.

Kemudian konfigurasi perangkat sesuai ketentuan

1. Mengkonfigurasi dasar perangkat : hostname, line console, enable password, banner message, disable domain IP lookup, encrypt all configured password. Sesuaikan hostname dengan ketentuan nama perangkat (Divis_Perangkat_NIU MAHASISWA, contoh : Management_SW_151192)

- Lakukan pada tiap access switch menyesuaikan nama departemen.

```
Management_SW_544609(config)#banner motd #this is Switch Management Lt1
Enter TEXT message.  End with the character '#'.

banner motd #this is Switch Management Lt1#

Management_SW_544609(config)#line console
% Incomplete command.
Management_SW_544609(config)#line console 0
Management_SW_544609(config-line)#pas
Management_SW_544609(config-line)#password cisco
Management_SW_544609(config-line)#login
Management_SW_544609(config-line)#ex
% Ambiguous command: "ex"
Management_SW_544609(config-line)#exit
Management_SW_544609(config)#line v
Management_SW_544609(config)#line vty 0 15
Management_SW_544609(config-line)#pass cisco
Management_SW_544609(config-line)#login
Management_SW_544609(config-line)#exit
Management_SW_544609(config)#no ip domain-
Management_SW_544609(config)#no ip domain-l
Management_SW_544609(config)#no ip domain-lookup
Management_SW_544609(config)#enable pass cisco
Management_SW_544609(config)#service pas
Management_SW_544609(config)#service password-encryption
Management_SW_544609(config)#do wr
Building configuration...
[OK]
```

Konfigurasi di Management Division Access Switch

```
Switch(config)#hostname Research_SW_544609
Research_SW_544609(config)#banner motd #this is Switch Research Lt
Enter TEXT message.  End with the character '#'.

banner motd #this is Switch Research Lt1#

Research_SW_544609(config)#line console 0
Research_SW_544609(config-line)#pass cisco
Research_SW_544609(config-line)#login
Research_SW_544609(config-line)#exit
Research_SW_544609(config)#line vty 0 15
Research_SW_544609(config-line)#pass cisco
Research_SW_544609(config-line)#login
Research_SW_544609(config-line)#exit
Research_SW_544609(config)#no ip domain-l
Research_SW_544609(config)#en pass cisco
% Ambiguous command: "en pass cisco"
Research_SW_544609(config)#enable pass cisco
Research_SW_544609(config)#ser
Research_SW_544609(config)#service
Research_SW_544609(config)#servicep
Research_SW_544609(config)#service p
Research_SW_544609(config)#service password-encryption
Research_SW_544609(config)#do wr
```

- Sama seperti pada switch access, setiap switch layer 3 dan router dilakukan juga konfigurasi dasar perangkat : hostname, line console, enable password, banner message, disable domain IP lookup, encrypt all configured password. Ditambah dengan konfigurasi SSH pada router dan L3 Switch. Sesuaikan hostname dengan ketentuan nama perangkat (LANTAI-NAMA PERANGKAT-NIU MAHASISWA, contoh : LT1_L3S_151192)

```
Switch(config)#hostname LT1_L3S_544609
LT1_L3S_544609(config)#bann
LT1_L3S_544609(config)#banner m
LT1_L3S_544609(config)#banner motd #this is Switch L3 Lt1#
LT1_L3S_544609(config)#line con 0
LT1_L3S_544609(config-line)#pass cisco
LT1_L3S_544609(config-line)#login
LT1_L3S_544609(config-line)#exit
LT1_L3S_544609(config)#
LT1_L3S_544609(config)#ip doma
LT1_L3S_544609(config)#ip domain-n
LT1_L3S_544609(config)#ip domain-name ubank.net
LT1_L3S_544609(config)#user
LT1_L3S_544609(config)#username ubank pass cisco
LT1_L3S_544609(config)#enc
LT1_L3S_544609(config)#cry
LT1_L3S_544609(config)#crypto ge
LT1_L3S_544609(config)#crypto gen
LT1_L3S_544609(config)#crypto key gen
LT1_L3S_544609(config)#crypto key generate rsa
The name for the keys will be: LT1_L3S_544609.ubank.net
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

LT1_L3S_544609(config)#line vty 0 15
*Mar 1 0:49:30.74: %SSH-5-ENABLED: SSH 1.99 has been enabled
LT1_L3S_544609(config-line)#login local
LT1_L3S_544609(config-line)#transport input ssh
LT1_L3S_544609(config-line)#exit
LT1_L3S_544609(config)#no ip domain-l
LT1_L3S_544609(config)#ena
LT1_L3S_544609(config)#enable pass cisco
LT1_L3S_544609(config)#se
LT1_L3S_544609(config)#service p
LT1_L3S_544609(config)#service password-encryption
LT1_L3S_544609(config)#do wr
```

Konfigurasi pada switch L3 (distribution switch)

Lakukan pada L3 yang lain dan juga setiap router.

2. Konfigurasi interface VLAN pada switch.

- Pada access switch, port yang mengarah ke distribution dibuat trunk, sementara port yang menuju end device dibuat access. Selain itu, dilakukan juga konfigurasi untuk switchportsecurity untuk access device. Lakukan pada setiap access switch

```
Management_SW_544609(config)#int range fa0/1-2
Management_SW_544609(config-if-range)#swit
Management_SW_544609(config-if-range)#switchport mode tr

Management_SW_544609(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/

Management_SW_544609(config-if-range)#ex
Management_SW_544609(config)#vlan 10
Management_SW_544609(config-vlan)#name management
Management_SW_544609(config-vlan)#ex
Management_SW_544609(config)#int range fa0/3-24
Management_SW_544609(config-if-range)#sw
Management_SW_544609(config-if-range)#switchport acc vlan 10
Management_SW_544609(config-if-range)#sw port max 2
Management_SW_544609(config-if-range)#sw port mac st
Management_SW_544609(config-if-range)#sw port vio sh
Management_SW_544609(config-if-range)#do wr
Building configuration...
[OK]
```

Contoh konfigurasi pada salah satu access switch (Management) di lt.1

- Pada distribution switch, interface port adalah trunk. Konfigurasikan pada setiap L3 Switch/

```
LT3_L3S_544609(config)#int range g1/0/2-7
LT3_L3S_544609(config-if-range)#sw tr enc d
% Interface range command failed for GigabitEthernet1/0/2
% Command failed on interface GigabitEthernet1/0/2. Aborting
LT3_L3S_544609(config)#int range g1/0/2-7
LT3_L3S_544609(config-if-range)#sw mode tr
LT3_L3S_544609(config-if-range)#end
```

Contoh konfigurasi pada switch distribution lt.3

3. Subnetting dan IP Addressing. Ketentuan awal adalah diberikan jaringan 192.168.10.0 sebagai alamat basis untuk jaringan dengan ketentuan, tiap departemen diperkirakan memiliki pengguna sekitar 60 orang, kecuali server room.

- Untuk pembagian jaringan itu, dilakukan subnetting. Karena pengguna diperkirakan sebanyak 60 orang, maka subnet mask yang sesuai adalah /26 = 255.255.255.255.192.

Tabel Hasil pembagian jaringan

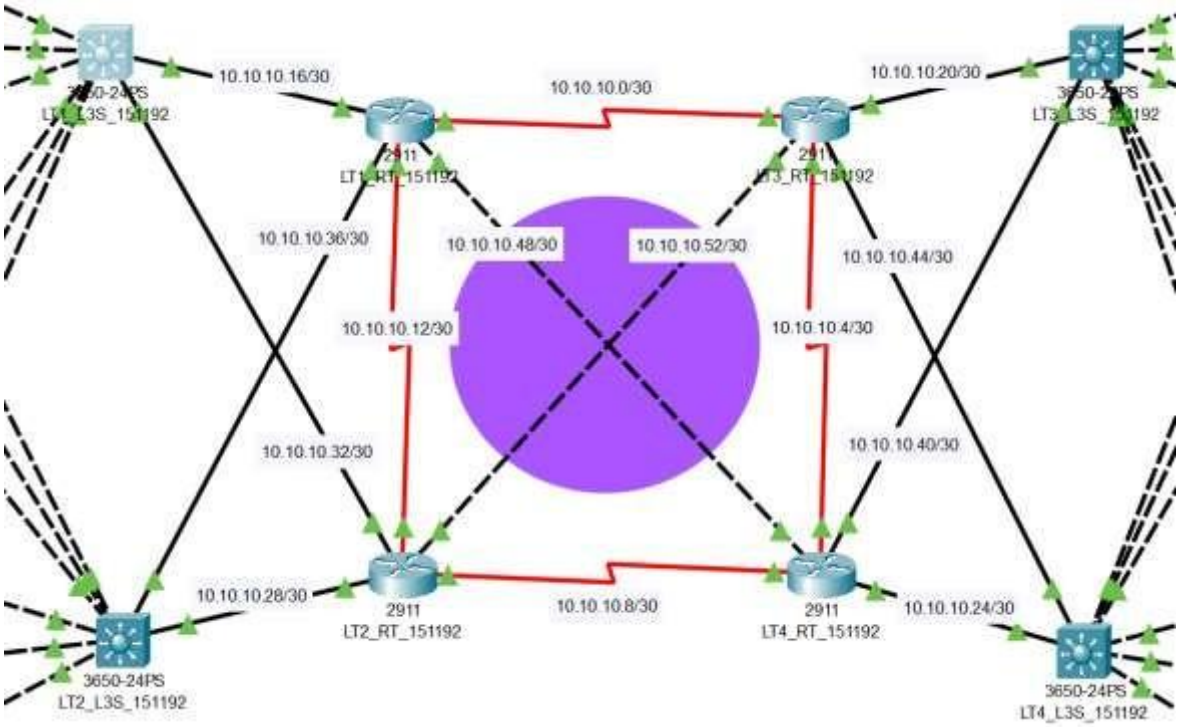
Department	Network Address	Subnet Mask	Host Address	Broadcast
Management	192.168.10.0	255.255.255.192	192.168.10.1 to 192.168.10.62	192.168.10.63
Research	192.168.10.64	255.255.255.192	192.168.10.65 to 192.168.10.126	192.168.10.127
HR	192.168.10.128	255.255.255.192	192.168.10.129 to 192.168.10.190	192.168.10.191
Marketing	192.168.10.192	255.255.255.192	192.168.10.193 to 192.168.10.254	192.168.10.255
Account	192.168.11.0	255.255.255.192	192.168.11.1 to 192.168.11.62	192.168.11.63
Finance	192.168.11.64	255.255.255.192	192.168.11.65 to 192.168.11.126	192.168.10.127
Logistic	192.168.11.128	255.255.255.192	192.168.11.129 to 192.168.11.190	192.168.11.191

Customer	192.168.11.192	255.255.255.192	192.168.11.193 to 192.168.11.254	192.168.11.255

Lanjutan Tabel Hasil pembagian jaringan

Department	Network Address	Subnet Mask	Host Address	Broadcast
Guest	192.168.12.0	255.255.255.192	192.168.12.1 to 192.168.12.62	192.168.12.63
Admin	192.168.12.64	255.255.255.192	192.168.12.65 to 192.168.12.126	192.168.12.127
ICT	192.168.12.128	255.255.255.192	192.168.12.129 to 192.168.12.190	192.168.12.191
Server Room	192.168.12.192	255.255.255.192	192.168.12.193 to 192.168.12.254	192.168.12.255

- Berikan alamat interface pada tiap perangkat. Berikut adalah detail IP address antar router dan switch.



Network Address antar Router dan Distribution Switch

- Pada switch distribution, berikan interface IP pada port yang mengarah ke router.

```
LT1_L3S_544609(config)#int r gl/0/1 , gl/0/8
LT1_L3S_544609(config-if-range)#no sw
LT1_L3S_544609(config-if-range)#no switchport
LT1_L3S_544609(config-if-range)#int gl/0/1
LT1_L3S_544609(config-if)#ip add 10.10.10.17 255.255.255.252
LT1_L3S_544609(config-if)#int gl/0/8
LT1_L3S_544609(config-if)#ip add 10.10.10.33 255.255.255.252
```

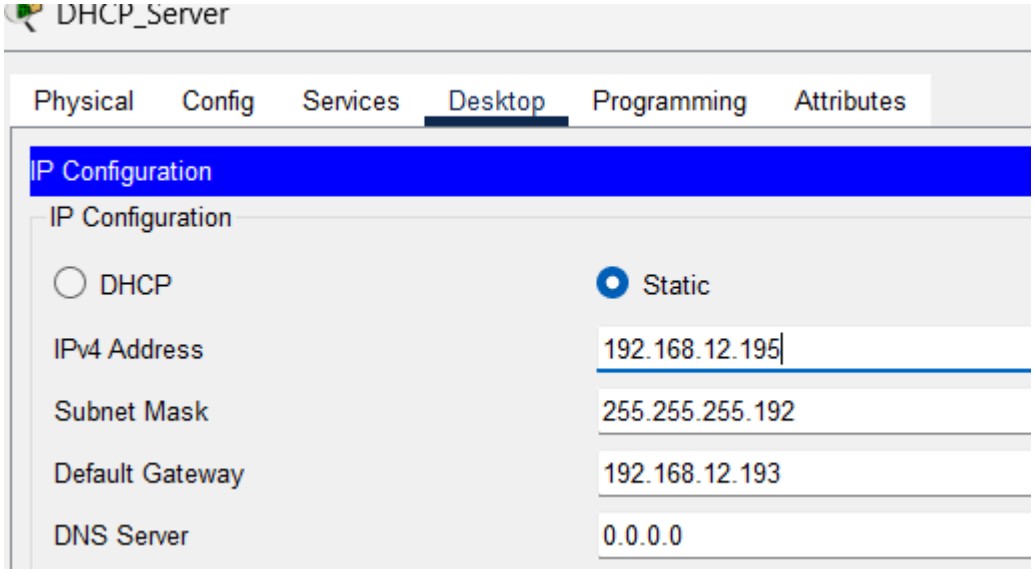
Contoh konfigurasi pada switch distribution lt.1

- Sementara itu pada tiap perangkat router, berikan interface IP pada seluruh portnya.
- Tambahkan clock rate pada port serial.

```
Router(config)#int g0/1
Router(config-if)#
Router(config-if)#ip add 10.10.10.18 255.255.255.252
Router(config-if)#int g0/2
Router(config-if)#ip add 10.10.10.38 255.255.255.252
Router(config-if)#int se0/3/1
Router(config-if)#ip add 10.10.10.1 255.255.255.252
Router(config-if)#int se0/3/0
Router(config-if)#ip add 10.10.10.13 255.255.255.252
Router(config-if)#int g0/0
Router(config-if)#ip add 10.10.10.49 255.255.255.252
Router(config-if)#do wr
Building configuration...
```

Contoh konfigurasi pada router lt.1

- Khusus pada server room, beri pengalamatan secara statis pada tiap perangkat server.



Contoh pengalamatan secara statis pada DHCP server

4. Routing menggunakan protokol OSPF.

- Pada tiap router, konfigurasikan routing OSPF dengan melampirkan alamat network yang terhubung dengan perangkat tersebut dan wildcard subnet mask nya.

```
Router(config)#router ospf 10
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 10.10.10.12 0.0.0.3 area 0
Router(config-router)#network 10.10.10.16 0.0.0.3 area 0
Router(config-router)#network 10.10.10.36 0.0.0.3 area 0
Router(config-router)#network 10.10.10.48 0.0.0.3 area 0
```

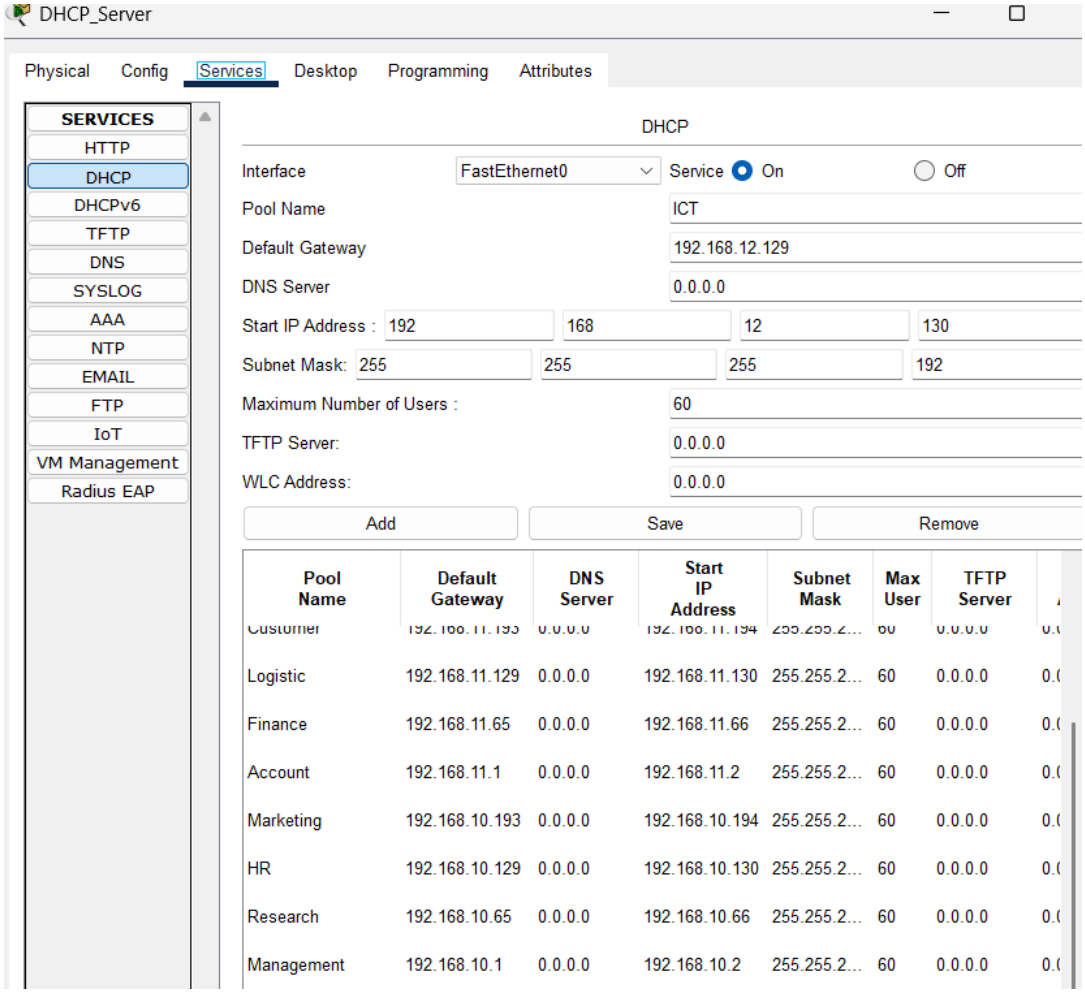
Contoh konfigurasi OSPF pada router lt.1

- Sama seperti router, pada tiap L3 Switch lakukan juga konfigurasi untuk routing. Pada switch, alamat network yang dilampirkan adalah alamat end network dibawahnya ditambah alamat yang mengarah ke router. Contohnya seperti berikut.

```
LT1_L3S_544609(config)#ip routing
LT1_L3S_544609(config)#router ospf 10
LT1_L3S_544609(config-router)#network 10.10.10.28 0.0.0.3 area 0
LT1_L3S_544609(config-router)#network 10.10.10.36 0.0.0.3 area 0
LT1_L3S_544609(config-router)#network 192.168.10.0 0.0.0.63 area 0
LT1_L3S_544609(config-router)#network 192.168.10.64 0.0.0.63 area 0
LT1_L3S_544609(config-router)#network 192.168.10.128 0.0.0.63 area 0
LT1_L3S_544609(config-router)#network 192.168.10.192 0.0.0.63 area 0
LT1_L3S_544609(config-router)#network 192.168.11.0 0.0.0.63 area 0
LT1_L3S_544609(config-router)#network 192.168.11.64 0.0.0.63 area 0
```

Contoh konfigurasi pada distribution switch lt.2

5. Konfigurasi perangkat DHCP Server. Buka tab Service pada perangkat dan pilih DHCP. Tambahkan Pool Name sesuai departemen dan masukkan gateway, subnet, dan start IP address sesuai pembagian alokasi jaringan sebelumnya, lalu klik Add. Lakukan secara berulang untuk tiap departemen (kecuali departemen server room).



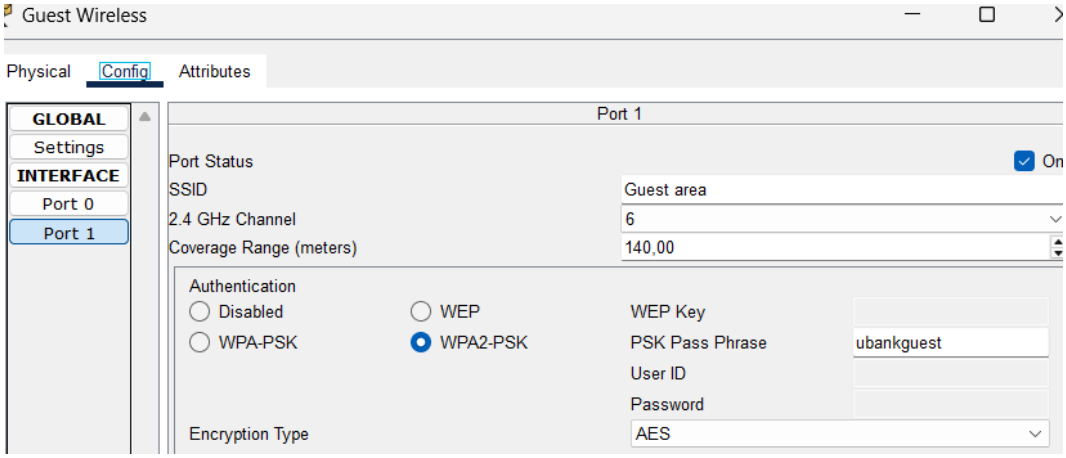
6. Konfigurasi inter-VLAN routing pada L3 Switch.

- Lakukan pada masing-masing L3 Switch dengan menyesuaikan IP address interface tiap departemen sesuai pengalamatan sebelumnya. Dilakukan juga konfigurasi untuk dhcp helper agar perangkat dibawahnya mendapatkan IP secara DHCP dengan menggunakan command *ip helper-address [ip address DHCP server]*.
- Namun terkhusus pada L3 Switch lantai 4 dimana terdapat departemen server room, saat mendaftarkan interface VLAN ruangan server room tidak perlu menggunakan command *ip helper-address*.

```
LT1_L3S_544609(config)#int vlan 10
LT1_L3S_544609(config-if)#ip add 192.168.10.1 255.255.255.192
LT1_L3S_544609(config-if)#no sh
LT1_L3S_544609(config-if)#ip help
LT1_L3S_544609(config-if)#ip helper-address 192.168.12.195
LT1_L3S_544609(config-if)#int vlan 20
LT1_L3S_544609(config-if)#ip add 192.168.10.65 255.255.255.192
LT1_L3S_544609(config-if)#no sh
LT1_L3S_544609(config-if)#ip helper-address 192.168.12.195
LT1_L3S_544609(config-if)#int vlan 30
LT1_L3S_544609(config-if)#ip add 192.168.10.129 255.255.255.192
LT1_L3S_544609(config-if)#no sh
LT1_L3S_544609(config-if)#ip helper-address 192.168.12.195
LT1_L3S_544609(config-if)#int vlan 40
LT1_L3S_544609(config-if)#ip add 192.168.10.193 255.255.255.192
LT1_L3S_544609(config-if)#no sh
LT1_L3S_544609(config-if)#ip helper-address 192.168.12.195
LT1_L3S_544609(config-if)#int vlan 50
LT1_L3S_544609(config-if)#ip add 192.168.11.1 255.255.255.192
LT1_L3S_544609(config-if)#no sh
LT1_L3S_544609(config-if)#ip helper-address 192.168.12.195
LT1_L3S_544609(config-if)#int vlan 60
LT1_L3S_544609(config-if)#ip add 192.168.11.65 255.255.255.192
LT1_L3S_544609(config-if)#no sh
LT1_L3S_544609(config-if)#ip helper-address 192.168.12.195
LT1_L3S_544609(config-if)#do wr
```

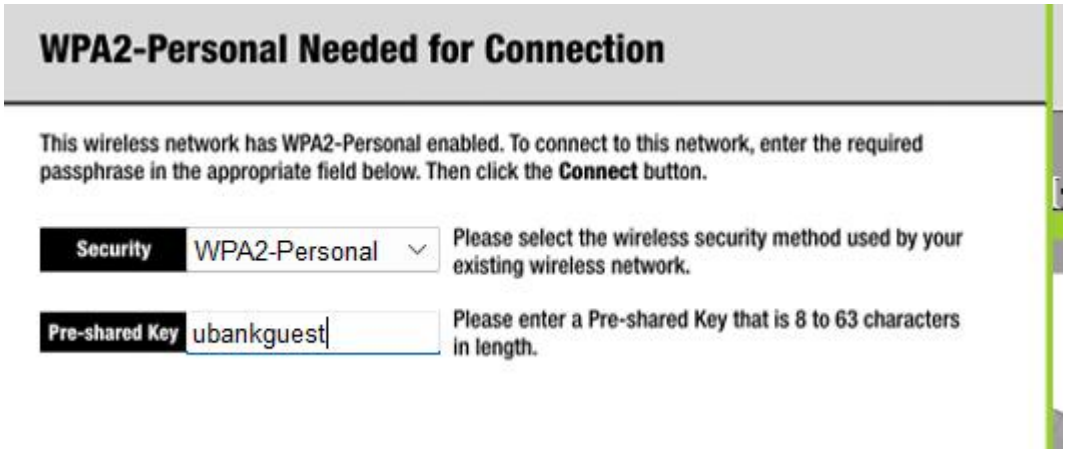
Contoh konfigurasi pada distribution switch lantai 1

7. Konfigurasi wireless network.



Pada access point di departemen guest, tambahkan SSID dan password dengan autentikasi WPA2-PSK

- Pada sisi laptop, tambahkan modul WPC300N, kemudian hubungkan ke wireless network pada menu Desktop > PC Wireless Pilih jaringan dan masukkan pass phrase.



8. Konfigurasi web server. Pada Web Server,aktifkan HTTP Service dan edit file index.html (kreasikan isinya, masukkan NIM di isian web), kemudian klik Save.

Physical Config Services Desktop Programming Attributes

SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP

File Name: index.html

```
space-between; align-items: center; }
  .topo-card-header span { font-size: 0.85rem; font-weight: 600; color: #2d3748; }
  .badge { font-size: 0.7rem; font-weight: 600; padding: 0.2rem 0.7rem; border-radius: 20px; background:
#ebf8ff; color: #2b6cb0; }
  .topo-img-wrap { padding: 1.5rem; background: #f7fafc; text-align: center; }
  .topo-img-wrap img { max-width: 100%; height: auto; border-radius: 8px; border: 1px solid #e2e8f0; }
  footer { background: white; border-top: 1px solid #e2e8f0; padding: 1.2rem 3rem; display: flex; justify-
content: space-between; align-items: center; flex-wrap: wrap; gap: 0.5rem; }
  .footer-left { font-size: 0.8rem; color: #4a5568; }
  .footer-left strong { color: #1a202c; }
  .footer-right { font-size: 0.75rem; color: #a0aec0; }
  @media (max-width: 600px) { header { padding: 1.5rem; gap: 1rem; } .student-bar { padding: 0.7rem 1.5rem;
gap: 1rem; } main { padding: 0 1rem 2rem; } footer { padding: 1rem 1.5rem; } }
</style>
</head>
<body>

<header>
  <div class="bank-logo">UB</div>
  <div class="header-text">
    <h1>U Bank</h1>
    <p>Perancangan dan Desain Jaringan  ·  Praktikum Perancangan Jaringan</p>
  </div>
</header>

<div class="student-bar">
  <span>Hilman Amirul Haq</span>
  <span>NIM: 24/544609/SV/25472</span>
  <span>Kelas RI4B1</span>
  <span>D4 Teknologi Rekayasa Internet – Sekolah Vokasi UGM</span>
</div>

<main>
  <div class="section-title">Topologi Jaringan</div>
  <div class="topo-card">
    <div class="topo-card-header">
      <span>Network Topology – U Bank</span>
      <span class="badge">Cisco Packet Tracer</span>
    </div>
    <div class="topo-img-wrap">
```

Web Browser

< > URL http://192.168.12.194 Go Stop

UB

U Bank

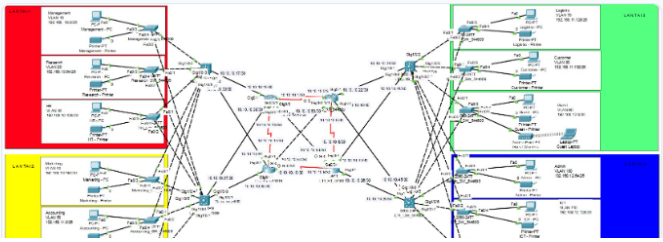
Perancangan dan Desain Jaringan · Praktikum Perancangan Jaringan

• Hilman Amirul Haq • NIM: 24/544609/SV/25472 • Kelas RI4B1

• D4 Teknologi Rekayasa Internet – Sekolah Vokasi UGM

TOPOLOGI JARINGAN

Network Topology – U Bank Cisco Packet Tracer



Checklist Pengujian Konfigurasi Topologi

1. Cek interface pada tiap perangkat. Screenshot pada masing-masing L3 Switch dan router.

```
LT1_L3S_544609#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   10.10.10.17     YES manual up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   10.10.10.33     YES manual up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.1    YES manual up                    up
Vlan20                 192.168.10.65   YES manual up                    up
Vlan30                 192.168.10.129  YES manual up                    up
Vlan40                 192.168.10.193  YES manual up                    up
Vlan50                 192.168.11.1    YES manual up                    up
Vlan60                 192.168.11.65   YES manual up                    up
```

```
LT2_L3S_544609#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   10.10.10.29     YES manual up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   10.10.10.37     YES manual up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.1    YES manual up                    up
Vlan20                 192.168.10.65   YES manual up                    up
Vlan30                 192.168.10.129  YES manual up                    up
Vlan40                 192.168.10.193  YES manual up                    up
Vlan50                 192.168.11.1    YES manual up                    up
Vlan60                 192.168.11.65   YES manual up                    up
```

```
LT3_L3S_544609#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   10.10.10.21     YES manual up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   10.10.10.41     YES manual up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan70                 192.168.11.129  YES manual up                    up
Vlan80                 192.168.11.193  YES manual up                    up
Vlan90                 192.168.12.1    YES manual up                    up
Vlan100                192.168.12.65   YES manual up                    up
Vlan110                192.168.12.129  YES manual up                    up
Vlan120                192.168.12.193  YES manual up                    up
LT4_L3S_544609#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   10.10.10.25     YES manual up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   10.10.10.45     YES manual up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan70                 192.168.11.129  YES manual up                    up
Vlan80                 192.168.11.193  YES manual up                    up
Vlan90                 192.168.12.1    YES manual up                    up
Vlan100                192.168.12.65   YES manual up                    up
Vlan110                192.168.12.129  YES manual up                    up
Vlan120                192.168.12.193  YES manual up                    up
```

```
LT1_RT_544609#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.49     YES manual up                    up
GigabitEthernet0/1     10.10.10.18     YES manual up                    up
GigabitEthernet0/2     10.10.10.38     YES manual up                    up
Serial0/3/0            10.10.10.13     YES manual up                    up
Serial0/3/1            10.10.10.1      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
LT2_RT_544609#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.54     YES manual up                    up
GigabitEthernet0/1     10.10.10.30     YES manual up                    up
GigabitEthernet0/2     10.10.10.34     YES manual up                    up
Serial0/3/0            10.10.10.10     YES manual up                    up
Serial0/3/1            10.10.10.14     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
LT3_RT_544609#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.53     YES manual up                    up
GigabitEthernet0/1     10.10.10.22     YES manual up                    up
GigabitEthernet0/2     10.10.10.46     YES manual up                    up
Serial0/3/0            10.10.10.2      YES manual up                    up
Serial0/3/1            10.10.10.5      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
LT4_RT_544609#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.50     YES manual up                    up
GigabitEthernet0/1     10.10.10.26     YES manual up                    up
GigabitEthernet0/2     10.10.10.42     YES manual up                    up
Serial0/3/0            10.10.10.6      YES manual up                    up
Serial0/3/1            10.10.10.9      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

2. Cek routing pada tiap router dan L3 Switch.

```
LT1_RT_544609#sh ip ro os
     10.0.0.0/8 is variably subnetted, 19 subnets, 2 masks
O       10.10.10.4 [110/65] via 10.10.10.50, 06:02:02, GigabitEthernet0/0
O       10.10.10.8 [110/65] via 10.10.10.50, 06:02:02, GigabitEthernet0/0
O       10.10.10.20 [110/3] via 10.10.10.50, 05:36:05, GigabitEthernet0/0
O       10.10.10.24 [110/2] via 10.10.10.50, 05:26:15, GigabitEthernet0/0
O       10.10.10.28 [110/2] via 10.10.10.37, 05:40:54, GigabitEthernet0/2
O       10.10.10.32 [110/2] via 10.10.10.17, 05:30:19, GigabitEthernet0/1
O       10.10.10.40 [110/2] via 10.10.10.50, 05:36:05, GigabitEthernet0/0
O       10.10.10.44 [110/3] via 10.10.10.50, 05:26:15, GigabitEthernet0/0
O       10.10.10.52 [110/3] via 10.10.10.17, 03:18:44, GigabitEthernet0/1
                    [110/3] via 10.10.10.37, 03:18:44, GigabitEthernet0/2
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/2] via 10.10.10.17, 00:57:04, GigabitEthernet0/1
                     [110/2] via 10.10.10.37, 00:57:04, GigabitEthernet0/2
O       192.168.10.64 [110/2] via 10.10.10.17, 00:57:04, GigabitEthernet0/1
                      [110/2] via 10.10.10.37, 00:57:04, GigabitEthernet0/2
O       192.168.10.128 [110/2] via 10.10.10.17, 00:57:04, GigabitEthernet0/1
                       [110/2] via 10.10.10.37, 00:57:04, GigabitEthernet0/2
O       192.168.10.192 [110/2] via 10.10.10.17, 00:57:04, GigabitEthernet0/1
                       [110/2] via 10.10.10.37, 00:57:04, GigabitEthernet0/2
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.0 [110/2] via 10.10.10.17, 00:57:04, GigabitEthernet0/1
                     [110/2] via 10.10.10.37, 00:57:04, GigabitEthernet0/2
O       192.168.11.64 [110/2] via 10.10.10.17, 00:57:04, GigabitEthernet0/1
                      [110/2] via 10.10.10.37, 00:57:04, GigabitEthernet0/2
O       192.168.11.128 [110/3] via 10.10.10.50, 00:05:32, GigabitEthernet0/0
O       192.168.11.192 [110/3] via 10.10.10.50, 00:05:32, GigabitEthernet0/0
     192.168.12.0/26 is subnetted, 4 subnets
O       192.168.12.0 [110/3] via 10.10.10.50, 00:05:32, GigabitEthernet0/0
O       192.168.12.64 [110/3] via 10.10.10.50, 00:05:32, GigabitEthernet0/0
O       192.168.12.128 [110/3] via 10.10.10.50, 00:05:05, GigabitEthernet0/0
O       192.168.12.192 [110/3] via 10.10.10.50, 00:05:32, GigabitEthernet0/0

LT2_RT_544609#sh ip ro osp
     10.0.0.0/8 is variably subnetted, 19 subnets, 2 masks
O       10.10.10.0 [110/65] via 10.10.10.53, 05:59:19, GigabitEthernet0/0
O       10.10.10.4 [110/65] via 10.10.10.53, 05:59:19, GigabitEthernet0/0
O       10.10.10.16 [110/2] via 10.10.10.33, 05:31:22, GigabitEthernet0/2
O       10.10.10.20 [110/2] via 10.10.10.53, 05:37:16, GigabitEthernet0/0
O       10.10.10.24 [110/3] via 10.10.10.53, 05:27:18, GigabitEthernet0/0
O       10.10.10.36 [110/2] via 10.10.10.29, 05:42:00, GigabitEthernet0/1
O       10.10.10.40 [110/3] via 10.10.10.53, 05:37:06, GigabitEthernet0/0
O       10.10.10.44 [110/2] via 10.10.10.53, 05:27:18, GigabitEthernet0/0
O       10.10.10.48 [110/3] via 10.10.10.29, 04:31:03, GigabitEthernet0/1
                    [110/3] via 10.10.10.33, 04:31:03, GigabitEthernet0/2
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/2] via 10.10.10.29, 00:58:07, GigabitEthernet0/1
                     [110/2] via 10.10.10.33, 00:58:07, GigabitEthernet0/2
O       192.168.10.64 [110/2] via 10.10.10.29, 00:58:07, GigabitEthernet0/1
                      [110/2] via 10.10.10.33, 00:58:07, GigabitEthernet0/2
O       192.168.10.128 [110/2] via 10.10.10.29, 00:58:07, GigabitEthernet0/1
                       [110/2] via 10.10.10.33, 00:58:07, GigabitEthernet0/2
O       192.168.10.192 [110/2] via 10.10.10.29, 00:58:07, GigabitEthernet0/1
                       [110/2] via 10.10.10.33, 00:58:07, GigabitEthernet0/2
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.0 [110/2] via 10.10.10.29, 00:58:07, GigabitEthernet0/1
                     [110/2] via 10.10.10.33, 00:58:07, GigabitEthernet0/2
O       192.168.11.64 [110/2] via 10.10.10.29, 00:58:07, GigabitEthernet0/1
                      [110/2] via 10.10.10.33, 00:58:07, GigabitEthernet0/2
O       192.168.11.128 [110/3] via 10.10.10.53, 00:55:11, GigabitEthernet0/0
O       192.168.11.192 [110/3] via 10.10.10.53, 00:53:09, GigabitEthernet0/0
     192.168.12.0/26 is subnetted, 4 subnets
O       192.168.12.0 [110/3] via 10.10.10.53, 00:53:09, GigabitEthernet0/0
O       192.168.12.64 [110/3] via 10.10.10.53, 00:53:09, GigabitEthernet0/0
O       192.168.12.128 [110/3] via 10.10.10.53, 00:53:09, GigabitEthernet0/0
O       192.168.12.192 [110/3] via 10.10.10.53, 00:53:09, GigabitEthernet0/0
```

```
LT3_RT_544609#sh ip ro os
     10.0.0.0/8 is variably subnetted, 19 subnets, 2 masks
O       10.10.10.8 [110/65] via 10.10.10.54, 03:20:10, GigabitEthernet0/0
O       10.10.10.12 [110/65] via 10.10.10.54, 03:20:10, GigabitEthernet0/0
O       10.10.10.16 [110/3] via 10.10.10.54, 03:20:10, GigabitEthernet0/0
O       10.10.10.24 [110/2] via 10.10.10.45, 05:27:43, GigabitEthernet0/2
O       10.10.10.28 [110/2] via 10.10.10.54, 03:20:10, GigabitEthernet0/0
O       10.10.10.32 [110/2] via 10.10.10.54, 03:20:10, GigabitEthernet0/0
O       10.10.10.36 [110/3] via 10.10.10.54, 03:20:10, GigabitEthernet0/0
O       10.10.10.40 [110/2] via 10.10.10.21, 05:37:34, GigabitEthernet0/1
O       10.10.10.48 [110/3] via 10.10.10.21, 04:31:36, GigabitEthernet0/1
                    [110/3] via 10.10.10.45, 04:31:36, GigabitEthernet0/2
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/3] via 10.10.10.54, 00:58:35, GigabitEthernet0/0
O       192.168.10.64 [110/3] via 10.10.10.54, 00:58:35, GigabitEthernet0/0
O       192.168.10.128 [110/3] via 10.10.10.54, 00:58:35, GigabitEthernet0/0
O       192.168.10.192 [110/3] via 10.10.10.54, 00:58:35, GigabitEthernet0/0
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.0 [110/3] via 10.10.10.54, 00:58:35, GigabitEthernet0/0
O       192.168.11.64 [110/3] via 10.10.10.54, 00:58:35, GigabitEthernet0/0
O       192.168.11.128 [110/2] via 10.10.10.21, 00:07:03, GigabitEthernet0/1
                       [110/2] via 10.10.10.45, 00:07:03, GigabitEthernet0/2
O       192.168.11.192 [110/2] via 10.10.10.21, 00:07:03, GigabitEthernet0/1
                       [110/2] via 10.10.10.45, 00:07:03, GigabitEthernet0/2
     192.168.12.0/26 is subnetted, 4 subnets
O       192.168.12.0 [110/2] via 10.10.10.21, 00:07:03, GigabitEthernet0/1
                     [110/2] via 10.10.10.45, 00:07:03, GigabitEthernet0/2
O       192.168.12.64 [110/2] via 10.10.10.21, 00:07:03, GigabitEthernet0/1
                      [110/2] via 10.10.10.45, 00:07:03, GigabitEthernet0/2
O       192.168.12.128 [110/2] via 10.10.10.21, 00:06:36, GigabitEthernet0/1
                       [110/2] via 10.10.10.45, 00:06:36, GigabitEthernet0/2
O       192.168.12.192 [110/2] via 10.10.10.21, 00:07:03, GigabitEthernet0/1
                       [110/2] via 10.10.10.45, 00:07:03, GigabitEthernet0/2
```

```
LT4_RT_544609#sh ip ro os
     10.0.0.0/8 is variably subnetted, 19 subnets, 2 masks
O       10.10.10.0 [110/65] via 10.10.10.49, 03:20:49, GigabitEthernet0/0
O       10.10.10.12 [110/65] via 10.10.10.49, 03:20:49, GigabitEthernet0/0
O       10.10.10.16 [110/2] via 10.10.10.49, 03:20:49, GigabitEthernet0/0
O       10.10.10.20 [110/2] via 10.10.10.41, 05:37:52, GigabitEthernet0/2
O       10.10.10.28 [110/3] via 10.10.10.49, 03:20:28, GigabitEthernet0/0
O       10.10.10.32 [110/3] via 10.10.10.49, 03:20:49, GigabitEthernet0/0
O       10.10.10.36 [110/2] via 10.10.10.49, 03:20:49, GigabitEthernet0/0
O       10.10.10.44 [110/2] via 10.10.10.25, 03:20:49, GigabitEthernet0/1
O       10.10.10.52 [110/3] via 10.10.10.25, 03:20:49, GigabitEthernet0/1
                    [110/3] via 10.10.10.41, 03:20:49, GigabitEthernet0/2
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/3] via 10.10.10.49, 01:02:58, GigabitEthernet0/0
O       192.168.10.64 [110/3] via 10.10.10.49, 01:02:31, GigabitEthernet0/0
O       192.168.10.128 [110/3] via 10.10.10.49, 01:02:31, GigabitEthernet0/0
O       192.168.10.192 [110/3] via 10.10.10.49, 01:02:31, GigabitEthernet0/0
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.0 [110/3] via 10.10.10.49, 01:02:31, GigabitEthernet0/0
O       192.168.11.64 [110/3] via 10.10.10.49, 01:02:31, GigabitEthernet0/0
O       192.168.11.128 [110/2] via 10.10.10.25, 00:07:21, GigabitEthernet0/1
                       [110/2] via 10.10.10.41, 00:07:21, GigabitEthernet0/2
O       192.168.11.192 [110/2] via 10.10.10.25, 00:07:21, GigabitEthernet0/1
                       [110/2] via 10.10.10.41, 00:07:21, GigabitEthernet0/2
     192.168.12.0/26 is subnetted, 4 subnets
O       192.168.12.0 [110/2] via 10.10.10.25, 00:07:21, GigabitEthernet0/1
                     [110/2] via 10.10.10.41, 00:07:21, GigabitEthernet0/2
O       192.168.12.64 [110/2] via 10.10.10.25, 00:07:21, GigabitEthernet0/1
                      [110/2] via 10.10.10.41, 00:07:21, GigabitEthernet0/2
O       192.168.12.128 [110/2] via 10.10.10.25, 00:06:54, GigabitEthernet0/1
                       [110/2] via 10.10.10.41, 00:06:54, GigabitEthernet0/2
O       192.168.12.192 [110/2] via 10.10.10.25, 00:07:21, GigabitEthernet0/1
                       [110/2] via 10.10.10.41, 00:07:21, GigabitEthernet0/2
```

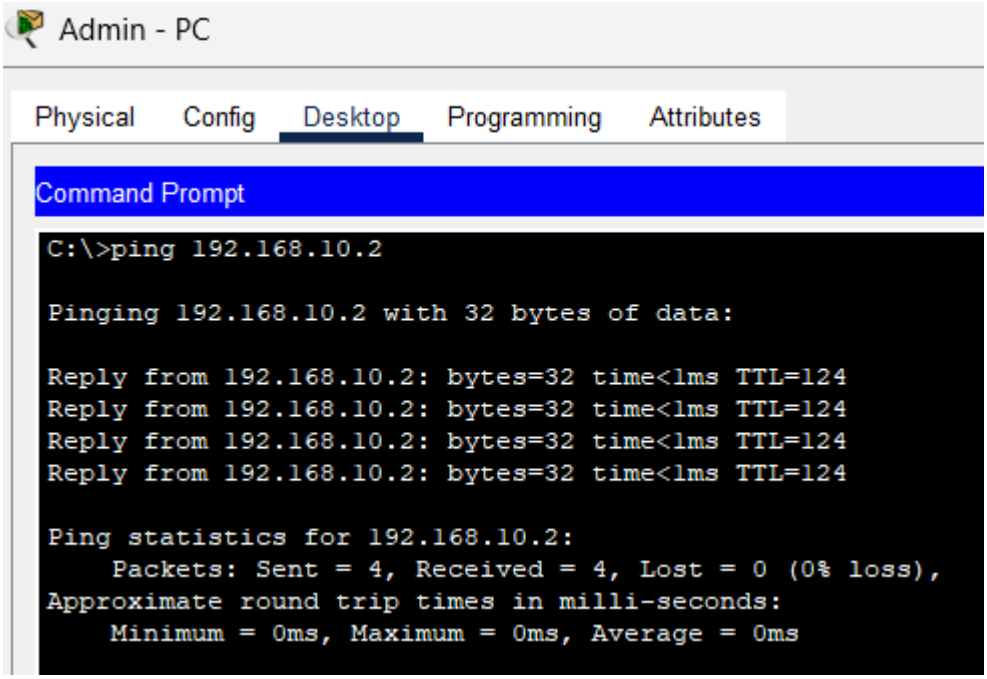
Contoh pada router

```
LT1_L3S_544609#sh ip ro os
     10.0.0.0/30 is subnetted, 14 subnets
O       10.10.10.0 [110/65] via 10.10.10.18, 05:32:32, GigabitEthernet1/0/1
O       10.10.10.4 [110/66] via 10.10.10.18, 05:32:32, GigabitEthernet1/0/1
                   [110/66] via 10.10.10.34, 05:32:32, GigabitEthernet1/0/8
O       10.10.10.8 [110/65] via 10.10.10.34, 05:32:32, GigabitEthernet1/0/8
O       10.10.10.12 [110/65] via 10.10.10.18, 05:32:32, GigabitEthernet1/0/1
                    [110/65] via 10.10.10.34, 05:32:32, GigabitEthernet1/0/8
O       10.10.10.20 [110/3] via 10.10.10.34, 05:32:32, GigabitEthernet1/0/8
O       10.10.10.24 [110/3] via 10.10.10.18, 05:28:28, GigabitEthernet1/0/1
O       10.10.10.28 [110/2] via 10.10.10.34, 05:32:32, GigabitEthernet1/0/8
O       10.10.10.36 [110/2] via 10.10.10.18, 05:32:32, GigabitEthernet1/0/1
O       10.10.10.40 [110/3] via 10.10.10.18, 05:32:32, GigabitEthernet1/0/1
O       10.10.10.44 [110/3] via 10.10.10.34, 05:28:28, GigabitEthernet1/0/8
O       10.10.10.48 [110/2] via 10.10.10.18, 04:32:13, GigabitEthernet1/0/1
O       10.10.10.52 [110/2] via 10.10.10.34, 04:32:13, GigabitEthernet1/0/8
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.128 [110/4] via 10.10.10.18, 00:07:45, GigabitEthernet1/0/1
                       [110/4] via 10.10.10.34, 00:07:45, GigabitEthernet1/0/8
O       192.168.11.192 [110/4] via 10.10.10.18, 00:07:45, GigabitEthernet1/0/1
                       [110/4] via 10.10.10.34, 00:07:45, GigabitEthernet1/0/8
     192.168.12.0/26 is subnetted, 4 subnets
O       192.168.12.0 [110/4] via 10.10.10.18, 00:07:45, GigabitEthernet1/0/1
                     [110/4] via 10.10.10.34, 00:07:45, GigabitEthernet1/0/8
O       192.168.12.64 [110/4] via 10.10.10.18, 00:07:45, GigabitEthernet1/0/1
                      [110/4] via 10.10.10.34, 00:07:45, GigabitEthernet1/0/8
O       192.168.12.128 [110/4] via 10.10.10.18, 00:07:18, GigabitEthernet1/0/1
                       [110/4] via 10.10.10.34, 00:07:18, GigabitEthernet1/0/8
O       192.168.12.192 [110/4] via 10.10.10.18, 00:07:45, GigabitEthernet1/0/1
                       [110/4] via 10.10.10.34, 00:07:45, GigabitEthernet1/0/8
LT2_L3S_544609#sh ip ro os
     10.0.0.0/30 is subnetted, 14 subnets
O       10.10.10.0 [110/65] via 10.10.10.38, 05:43:42, GigabitEthernet1/0/8
O       10.10.10.4 [110/66] via 10.10.10.38, 05:43:32, GigabitEthernet1/0/8
                   [110/66] via 10.10.10.30, 05:43:32, GigabitEthernet1/0/1
O       10.10.10.8 [110/65] via 10.10.10.30, 05:43:32, GigabitEthernet1/0/1
O       10.10.10.12 [110/65] via 10.10.10.38, 05:43:32, GigabitEthernet1/0/8
                    [110/65] via 10.10.10.30, 05:43:32, GigabitEthernet1/0/1
O       10.10.10.16 [110/2] via 10.10.10.38, 05:32:57, GigabitEthernet1/0/8
O       10.10.10.20 [110/3] via 10.10.10.30, 05:38:51, GigabitEthernet1/0/1
O       10.10.10.24 [110/3] via 10.10.10.38, 05:28:53, GigabitEthernet1/0/8
O       10.10.10.32 [110/2] via 10.10.10.30, 05:32:57, GigabitEthernet1/0/1
O       10.10.10.40 [110/3] via 10.10.10.38, 05:38:41, GigabitEthernet1/0/8
O       10.10.10.44 [110/3] via 10.10.10.30, 05:28:53, GigabitEthernet1/0/1
O       10.10.10.48 [110/2] via 10.10.10.38, 04:32:38, GigabitEthernet1/0/8
O       10.10.10.52 [110/2] via 10.10.10.30, 04:32:38, GigabitEthernet1/0/1
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.128 [110/4] via 10.10.10.38, 00:08:10, GigabitEthernet1/0/8
                       [110/4] via 10.10.10.30, 00:08:10, GigabitEthernet1/0/1
O       192.168.11.192 [110/4] via 10.10.10.38, 00:08:10, GigabitEthernet1/0/8
                       [110/4] via 10.10.10.30, 00:08:10, GigabitEthernet1/0/1
     192.168.12.0/26 is subnetted, 4 subnets
O       192.168.12.0 [110/4] via 10.10.10.38, 00:08:10, GigabitEthernet1/0/8
                     [110/4] via 10.10.10.30, 00:08:10, GigabitEthernet1/0/1
O       192.168.12.64 [110/4] via 10.10.10.38, 00:08:10, GigabitEthernet1/0/8
                      [110/4] via 10.10.10.30, 00:08:10, GigabitEthernet1/0/1
O       192.168.12.128 [110/4] via 10.10.10.38, 00:07:43, GigabitEthernet1/0/8
                       [110/4] via 10.10.10.30, 00:07:43, GigabitEthernet1/0/1
O       192.168.12.192 [110/4] via 10.10.10.38, 00:08:10, GigabitEthernet1/0/8
                       [110/4] via 10.10.10.30, 00:08:10, GigabitEthernet1/0/1
```

```
LT3_L3S_544609#sh ip ro os
     10.0.0.0/30 is subnetted, 14 subnets
O       10.10.10.0 [110/65] via 10.10.10.22, 03:21:36, GigabitEthernet1/0/1
O       10.10.10.4 [110/65] via 10.10.10.22, 03:21:36, GigabitEthernet1/0/1
                   [110/65] via 10.10.10.42, 03:21:36, GigabitEthernet1/0/8
O       10.10.10.8 [110/65] via 10.10.10.42, 03:21:36, GigabitEthernet1/0/8
O       10.10.10.12 [110/66] via 10.10.10.22, 03:21:36, GigabitEthernet1/0/1
                    [110/66] via 10.10.10.42, 03:21:36, GigabitEthernet1/0/8
O       10.10.10.16 [110/3] via 10.10.10.42, 03:21:36, GigabitEthernet1/0/8
O       10.10.10.24 [110/2] via 10.10.10.42, 03:21:36, GigabitEthernet1/0/8
O       10.10.10.28 [110/3] via 10.10.10.22, 03:21:36, GigabitEthernet1/0/1
O       10.10.10.32 [110/3] via 10.10.10.22, 03:21:36, GigabitEthernet1/0/1
O       10.10.10.36 [110/3] via 10.10.10.42, 03:21:36, GigabitEthernet1/0/8
O       10.10.10.44 [110/2] via 10.10.10.22, 03:21:36, GigabitEthernet1/0/1
O       10.10.10.48 [110/2] via 10.10.10.42, 03:21:36, GigabitEthernet1/0/8
O       10.10.10.52 [110/2] via 10.10.10.22, 03:21:36, GigabitEthernet1/0/1
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/4] via 10.10.10.22, 01:00:01, GigabitEthernet1/0/1
                     [110/4] via 10.10.10.42, 01:00:01, GigabitEthernet1/0/8
O       192.168.10.64 [110/4] via 10.10.10.22, 01:00:01, GigabitEthernet1/0/1
                      [110/4] via 10.10.10.42, 01:00:01, GigabitEthernet1/0/8
O       192.168.10.128 [110/4] via 10.10.10.22, 01:00:01, GigabitEthernet1/0/1
                       [110/4] via 10.10.10.42, 01:00:01, GigabitEthernet1/0/8
O       192.168.10.192 [110/4] via 10.10.10.22, 01:00:01, GigabitEthernet1/0/1
                       [110/4] via 10.10.10.42, 01:00:01, GigabitEthernet1/0/8
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.0 [110/4] via 10.10.10.22, 01:00:01, GigabitEthernet1/0/1
                     [110/4] via 10.10.10.42, 01:00:01, GigabitEthernet1/0/8
O       192.168.11.64 [110/4] via 10.10.10.22, 01:00:01, GigabitEthernet1/0/1
                      [110/4] via 10.10.10.42, 01:00:01, GigabitEthernet1/0/8
LT4_L3S_544609#sh ip ro os
     10.0.0.0/30 is subnetted, 14 subnets
O       10.10.10.0 [110/65] via 10.10.10.46, 05:29:32, GigabitEthernet1/0/8
O       10.10.10.4 [110/65] via 10.10.10.46, 05:29:22, GigabitEthernet1/0/8
                   [110/65] via 10.10.10.26, 05:29:22, GigabitEthernet1/0/1
O       10.10.10.8 [110/65] via 10.10.10.26, 05:29:22, GigabitEthernet1/0/1
O       10.10.10.12 [110/66] via 10.10.10.46, 05:29:22, GigabitEthernet1/0/8
                    [110/66] via 10.10.10.26, 05:29:22, GigabitEthernet1/0/1
O       10.10.10.16 [110/3] via 10.10.10.26, 05:29:22, GigabitEthernet1/0/1
O       10.10.10.20 [110/2] via 10.10.10.46, 05:29:32, GigabitEthernet1/0/8
O       10.10.10.28 [110/3] via 10.10.10.46, 05:29:32, GigabitEthernet1/0/8
O       10.10.10.32 [110/3] via 10.10.10.46, 05:29:32, GigabitEthernet1/0/8
O       10.10.10.36 [110/3] via 10.10.10.26, 05:29:22, GigabitEthernet1/0/1
O       10.10.10.40 [110/2] via 10.10.10.26, 05:29:22, GigabitEthernet1/0/1
O       10.10.10.48 [110/2] via 10.10.10.26, 04:33:15, GigabitEthernet1/0/1
O       10.10.10.52 [110/2] via 10.10.10.46, 04:33:15, GigabitEthernet1/0/8
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/4] via 10.10.10.46, 01:00:14, GigabitEthernet1/0/8
                     [110/4] via 10.10.10.26, 01:00:14, GigabitEthernet1/0/1
O       192.168.10.64 [110/4] via 10.10.10.46, 01:00:14, GigabitEthernet1/0/8
                      [110/4] via 10.10.10.26, 01:00:14, GigabitEthernet1/0/1
O       192.168.10.128 [110/4] via 10.10.10.46, 01:00:14, GigabitEthernet1/0/8
                       [110/4] via 10.10.10.26, 01:00:14, GigabitEthernet1/0/1
O       192.168.10.192 [110/4] via 10.10.10.46, 01:00:14, GigabitEthernet1/0/8
                       [110/4] via 10.10.10.26, 01:00:14, GigabitEthernet1/0/1
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.0 [110/4] via 10.10.10.46, 01:00:14, GigabitEthernet1/0/8
                     [110/4] via 10.10.10.26, 01:00:14, GigabitEthernet1/0/1
O       192.168.11.64 [110/4] via 10.10.10.46, 01:00:14, GigabitEthernet1/0/8
                      [110/4] via 10.10.10.26, 01:00:14, GigabitEthernet1/0/1
```

Contoh pada distribution switch

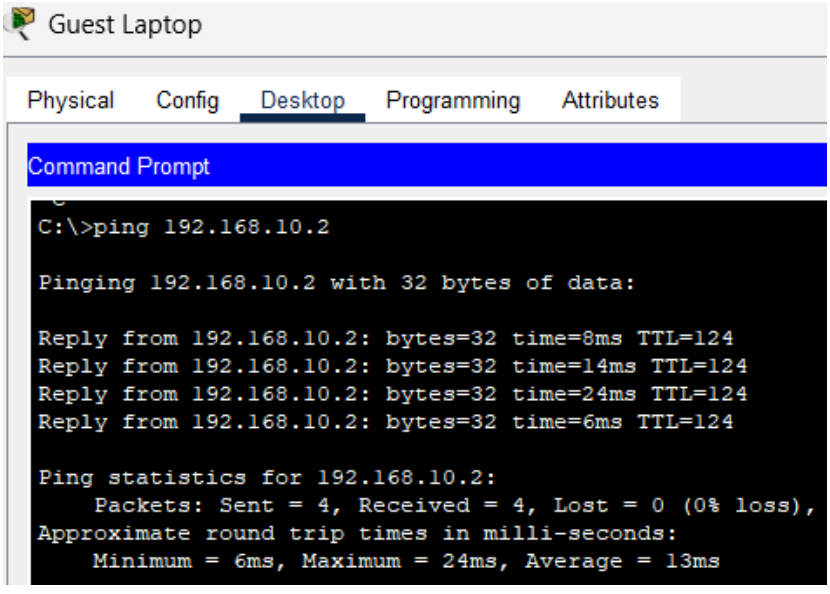
3. Cek koneksi. Ambil salah satu contoh PING antar perangkat.



PC pada departemen Admin dapat PING ke PC pada departemen Management

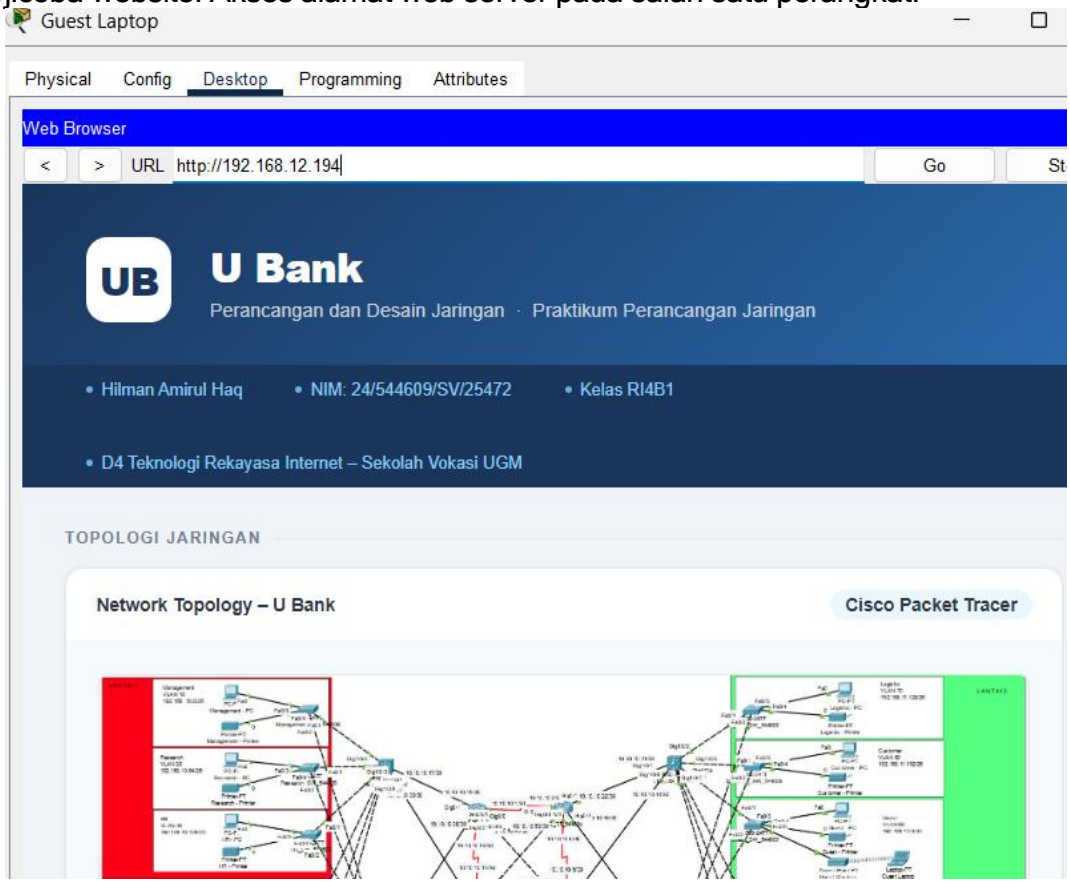
4. Cek Koneksi Perangkat Wireless. Ujicoba dengan PING ke PC di departemen lain

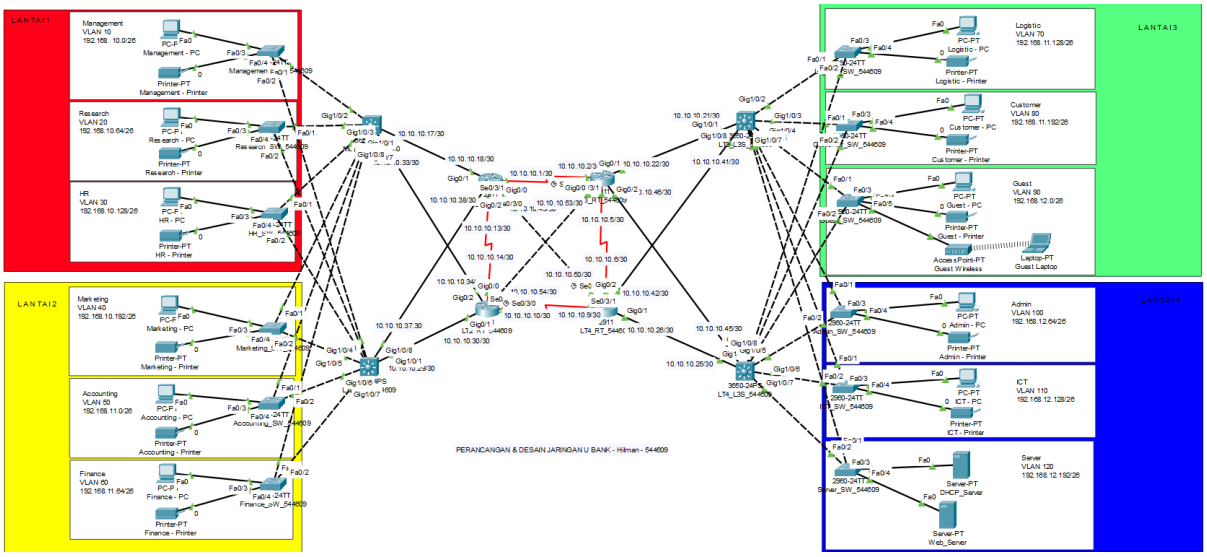




Laptop PING ke PC departemen management

5. Ujicoba website. Akses alamat web server pada salah satu perangkat.





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